

Rectangular Completeness Encompasses Standard Physical Closure

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Abstract

We prove that *closure as rectangular completeness*, developed in the comparison-world framework of [1], stands in a precise relation to the product-structure component of standard two-subsystem closure. For classical Hamiltonian systems, rectangular completeness is exactly equivalent to the full-product state-space condition, under separating cross-subsystem observables and the normalization that each subsystem state is realized by some joint state. The thermodynamic preparation-space analogue follows as a direct corollary. For quantum systems, the same comparison theorem holds on the product-state sector, while the full density-matrix state space contains correlation data not determined by subsystem marginal profiles. Because rectangular completeness is formulated purely in terms of binary comparison predicates, with no topology, metric, Hamiltonian, or dynamics presupposed, it extends the full-product component of physical closure to abstract relational comparison data. The result is a comparison theorem: it identifies the structural condition recovered from familiar two-subsystem formalisms, without replacing the separate dynamical requirements by which classical, thermodynamic, or quantum systems are usually called closed.

Keywords. Rectangular completeness, comparison worlds, closed systems, thermodynamic closure, product states, entanglement

1 Introduction

The notion of a *closed system* is used in several related but not identical senses across physics. In classical mechanics, one separates the state-space specification of

a composite system from the dynamical question whether its Hamiltonian couples to external degrees of freedom [3, Ch. 8]. In thermodynamics, a closed system exchanges no matter with its environment, while an isolated system exchanges neither matter nor energy [4, Chs. 1–2]. In quantum mechanics and quantum information theory, a closed system is usually represented by unitary evolution on the relevant Hilbert space, with no external-system coupling [5, § 2.2]. These uses are not interchangeable. They do, however, contain a common state-combination question: which pairs of subsystem profiles are available as states of the composite under consideration?

The paper [1] introduces a definition of closure in the abstract setting of *comparison worlds*. A comparison world $(U, \mathcal{C})$ consists of a set of joint states U and a family of binary predicates $\mathcal{C}$ on U ; no dynamics, topology, or background geometry is assumed. Closure is identified with *rectangular completeness*: every admissible pairing of a left-profile equivalence class and a right-profile equivalence class is realized by exactly one state. This condition forces a canonical product decomposition $U \cong X_A \times X_B$ and a diagonal symmetry-group action, from which quotient semantics [1, Thm. 3.2] and the failure of subsystem attribution [1, Thm. 7.1] are derived. This paper should therefore be read as a comparison theorem following [1] and the broader monograph [2]: it asks which part of standard two-subsystem closure is recovered by rectangular completeness.

The present paper answers the natural foundational question:

How does rectangular completeness relate to the product-structure component of standard physical closure? On which sectors is it equivalent, and in what sense is it more general as a relational condition?

The answer is twofold. For classical Hamiltonian systems equipped with their natural *cross-subsystem* comparison predicates, rectangular completeness is *exactly equivalent* to the full-product component of standard closure (Theorem 5.1). The thermodynamic equivalence follows as a corollary (Corollary 5.4), and the same equivalence holds for the product-state sector of quantum mechanics (Theorem 5.7). At the same time, rectangular completeness is *strictly more general* than those physical sectors: it applies to finite discrete comparison worlds with no dynamics whatsoever (Section 6). These results together show that rectangular completeness is the product-availability condition abstracted from standard two-subsystem formalisms.

The equivalence has non-trivial discriminatory content once the subsystem profiles have been fixed. A system with a kinematic constraint—such as two particles connected by a rigid rod—may be a closed composite mechanical system in the ordinary sense, but it is not closed relative to independently variable A - and B -profiles (Example 5.3). The missing profile pairs are exactly what rectangular completeness records. The thermodynamic analogue is the distinction between a preparation product and an equilibrium submanifold imposed by contact conditions (Example 5.6).

Organization. Section 2 recalls the three standard definitions of closure. Section 3 reviews the comparison-world machinery of [1]. Section 4 introduces cross-subsystem

comparison predicates and proves the profile-factorization lemma. Section 5 states and proves the main results. Section 6 establishes strict generality. Section 7 concludes.

2 Standard Notions of Physical Closure

We recall the standard physical terminology in enough detail to isolate the component compared with rectangular completeness. Throughout, a *two-subsystem system* consists of two distinguished components A and B .

2.1 Classical Hamiltonian systems

Let X_A and X_B be the state spaces (phase spaces) of subsystems A and B respectively.

Definition 2.1 (Classical product-and-nonexternality conditions [3, Ch. 8]). A two-subsystem classical Hamiltonian model has the closure form used here when:

- (i) the joint state space is the full Cartesian product, $U_{\text{cl}} = X_A \times X_B$; and
- (ii) the Hamiltonian $H: X_A \times X_B \rightarrow \mathbb{R}$ involves no coupling to any external degree of freedom— H is a function of the joint state alone, with no additional environment variable.

Condition (i) is the *full-product* condition: every combination of an A -state and a B -state is represented as a state of the composite model. Condition (ii) is the *nonexternality* condition: evolution is determined by the joint state and not by a further environment variable. Condition (i) is the operative one for the structural theorems of this paper; condition (ii) concerns dynamics, which the comparison-world framework does not assume.

Remark 2.2 (Missing profile pairs). A proper subset $U_{\text{eff}} \subsetneq X_A \times X_B$ may arise from several sources: an imposed kinematic constraint, a restriction to a sector, or an effective description after interaction with an environment. Rectangular completeness does not by itself identify the physical source of the restriction. It records the formal fact that some (α -class, β -class) pairs are unrealized.

2.2 Thermodynamic systems

Let X_A^{th} and X_B^{th} denote the spaces of thermodynamic states of components A and B respectively. A thermodynamic state records macrostate variables (internal energy, volume, particle numbers, etc.) sufficient to determine the component's thermodynamic properties.

Definition 2.3 (Standard thermodynamic closure [4, Ch. 1–2]). A two-component thermodynamic system is:

- *closed* if no matter is exchanged with the external environment;
- *isolated* if neither matter nor energy is exchanged with the external environment.

In the subsystem-level comparison setting considered here, the relevant state-combination question concerns the *preparation state space*—the set of independently preparable joint macrostates prior to any constrained equilibrium condition.

Remark 2.4 (Preparation versus equilibrium). The *preparation state space* (independently attainable initial macrostates) must be distinguished from an *equilibrium manifold* (states satisfying contact or extremality conditions, e.g. equal temperatures or pressures). Such an equilibrium manifold is generally a proper subset of the preparation product. The comparison-world construction is applied here to the preparation state space, which is the relevant full-product state space for the subsystem-level comparison developed in this paper, because the comparison predicates of Section 4 compare arbitrary joint states, not merely equilibrium states.

2.3 Quantum systems

Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be Hilbert spaces for subsystems A and B , with joint Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

Definition 2.5 (Standard quantum closure [5, § 2.2]). A bipartite quantum system is *standardly closed* if it evolves unitarily under a Hamiltonian H on $\mathcal{H}_A \otimes \mathcal{H}_B$ with no coupling to an external system.

The full state space $\mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ (density matrices) is not a Cartesian product of the individual-subsystem state spaces when entanglement is present. Section 5.3 shows that rectangular completeness captures the product-state sector, while marginal-profile comparison data do not determine the full state space.

3 Comparison Worlds and Rectangular Completeness

We recall the essential machinery from [1], where all proofs appear.

Definition 3.1 (Comparison world [1, Def. 2.1]). A *comparison world* is a pair $(U, \mathcal{C})$ where U is a nonempty set of states and $\mathcal{C}$ is a set of binary predicates $c: U \times U \rightarrow \{0, 1\}$, in the sense of relational structures [6]. No topology, metric, manifold structure, or background geometry is assumed.

For each $u \in U$ define the *left profile* $L(u): \mathcal{C} \times U \rightarrow \{0, 1\}$ by $L(u)(c, w) = c(u, w)$, and the *right profile* $R(u): \mathcal{C} \times U \rightarrow \{0, 1\}$ by $R(u)(c, w) = c(w, u)$. Define equivalence relations α and β on U by

$$u \alpha v \iff L(u) = L(v), \quad u \beta v \iff R(u) = R(v). \quad (1)$$

Set $X_A := U/\alpha$, $X_B := U/\beta$, and define the *canonical factor map*

$$\Theta: U \rightarrow X_A \times X_B, \quad \Theta(u) := ([u]_\alpha, [u]_\beta). \quad (2)$$

Definition 3.2 (Rectangular completeness [1, Def. 2.6]). The comparison world $(U, \mathcal{C})$ is *rectangularly complete* if for every $A \in X_A$ and every $B \in X_B$ there exists a *unique* $u \in U$ with $[u]_\alpha = A$ and $[u]_\beta = B$. Equivalently, the canonical factor map Θ is a bijection $U \xrightarrow{\sim} X_A \times X_B$.

Remark 3.3 (Internal reading). Rectangular completeness says there are no *missing* states and no *redundant* states relative to the product $X_A \times X_B$ determined by the comparison data alone. Each pair of profile classes is realized by exactly one state. This is the purely internal expression of full profile-pair availability.

The following results are proved in [1].

Theorem 3.4 (Diagonal action theorem [1, Thm. 2.8]). *If $(U, \mathcal{C})$ is rectangularly complete, then Θ is a bijection $U \cong X_A \times X_B$, and $G = \text{Aut}(U, \mathcal{C})$ acts diagonally on $X_A \times X_B$. The orbit quotient $\pi: X_A \times X_B \rightarrow \text{Phys} := (X_A \times X_B)/G$ is canonically determined.*

Theorem 3.5 (Minimality [1, Thm. 2.9]). *Rectangular completeness is the minimal internal condition that forces Θ to be bijective across all comparison worlds.*

Theorem 3.6 (Asymmetry [1, Cor. 2.11]). *If $(U, \mathcal{C})$ is rectangularly complete and $|U| > 1$, then $\mathcal{C}$ contains at least one asymmetric predicate: there exist $c \in \mathcal{C}$ and $u, w \in U$ with $c(u, w) \neq c(w, u)$.*

Theorem 3.6 shows that any genuinely two-subsystem comparison world must contain at least one asymmetric predicate. This motivates the class of predicates introduced in the next section.

4 Cross-Subsystem Comparison Predicates

The key to relating rectangular completeness to standard physical closure is the choice of comparison predicates. We introduce a canonical class that is both physically natural and structurally necessary.

Definition 4.1 (Observable separation). Let Z be a set and $\mathcal{F}$ a family of functions $f: Z \rightarrow \mathbb{R}$. We say $\mathcal{F}$ *separates* Z if for any $z \neq z'$ in Z there exists $f \in \mathcal{F}$ with $f(z) \neq f(z')$.

Remark 4.2 (Constant observables). Throughout we assume the observable families contain all constant functions $z \mapsto t$, $t \in \mathbb{R}$. This allows comparison against arbitrary scalar thresholds without affecting which subsystem states are separated or identified.

Definition 4.3 (Cross-subsystem predicate). Let $U = X_A \times X_B$, and let $\mathcal{F}_A$ (resp. $\mathcal{F}_B$) be a family of functions $X_A \rightarrow \mathbb{R}$ (resp. $X_B \rightarrow \mathbb{R}$). For each pair $(f, g) \in \mathcal{F}_A \times \mathcal{F}_B$, the *cross-subsystem comparison predicate* $c_{f,g}$ is

$$c_{f,g}((u_A, u_B), (v_A, v_B)) := \mathbf{1}[f(u_A) \geq g(v_B)]. \quad (3)$$

The family of all such predicates, $\mathcal{C}_{cs} := \{c_{f,g} : f \in \mathcal{F}_A, g \in \mathcal{F}_B\}$, is the *cross-subsystem comparison family* generated by $(\mathcal{F}_A, \mathcal{F}_B)$.

Remark 4.4 (Physical naturality). A cross-subsystem predicate $c_{f,g}$ compares an A -observable of the left state with a B -observable of the right state. Representative examples: the kinetic energy of A in state u versus the potential energy of B in state v ; the position of particle A along axis i in u versus the momentum of particle B along axis j in v ; the internal energy of thermodynamic component A in u versus the entropy of component B in v . These are the natural *relational comparisons between the two subsystems*: they record how the A -content of one state relates to the B -content of another, and are generically asymmetric—reversing the arguments exchanges the role of A and B observables with no algebraic reason for the values to agree.

Remark 4.5 (Why this family is canonical). The cross-subsystem family $\mathcal{C}_{cs}$ is canonical for three reasons: it is built directly from subsystem observables with no joint-system input; it compares the A -content of one state with the B -content of another, thereby resolving the two subsystem coordinates independently; and it supplies the asymmetry required by Theorem 3.6. It is the leanest observable-generated comparison language that accomplishes all three.

The next lemma is the structural key to all main results.

Lemma 4.6 (Profile factorization). *Let $U = Y_A \times Y_B$ and let $\mathcal{C} = \mathcal{C}_{cs}$ be the cross-subsystem family generated by separating families $\mathcal{F}_A$ on Y_A and $\mathcal{F}_B$ on Y_B , each containing all constant functions. Then the following hold.*

- (i) *The left profile $L(u)$ depends only on the A -component: if $u_A = v_A$, then $L(u) = L(v)$.*
- (ii) *The right profile $R(u)$ depends only on the B -component: if $u_B = v_B$, then $R(u) = R(v)$.*
- (iii) *The congruence α is determined by the A -component:*

$$(u_A, u_B) \alpha (v_A, v_B) \iff u_A \sim_{\mathcal{F}_A} v_A.$$

- (iv) *The congruence β is determined by the B -component:*

$$(u_A, u_B) \beta (v_A, v_B) \iff u_B \sim_{\mathcal{F}_B} v_B.$$

(v) Consequently, $U/\alpha \cong Y_A/\sim_{\mathcal{F}_A}$ and $U/\beta \cong Y_B/\sim_{\mathcal{F}_B}$. When $\mathcal{F}_A$ separates Y_A (and $\mathcal{F}_B$ separates Y_B), both quotients are trivial: $U/\alpha \cong Y_A$ and $U/\beta \cong Y_B$.

Proof. (i). For any $c_{f,g} \in \mathcal{C}_{cs}$ and $w \in U$, $c_{f,g}(u, w) = \mathbf{1}[f(u_A) \geq g(w_B)] = \mathbf{1}[f(v_A) \geq g(w_B)] = c_{f,g}(v, w)$, using $u_A = v_A$. Since $c_{f,g}$ was arbitrary, $L(u) = L(v)$.

(ii). Similarly, $c_{f,g}(w, u) = \mathbf{1}[f(w_A) \geq g(u_B)] = \mathbf{1}[f(w_A) \geq g(v_B)] = c_{f,g}(w, v)$ using $u_B = v_B$, so $R(u) = R(v)$.

(iii). $u \alpha v$ means $c_{f,g}(u, w) = c_{f,g}(v, w)$ for all f, g, w . Fix $f \in \mathcal{F}_A$; for each $t \in \mathbb{R}$ let g_t be the constant function with value t . Then $\mathbf{1}[f(u_A) \geq t] = c_{f,g_t}(u, w) = c_{f,g_t}(v, w) = \mathbf{1}[f(v_A) \geq t]$ for all t . If $f(u_A) \neq f(v_A)$, choose t strictly between the two values; the two indicator functions then differ at t , a contradiction. Hence $f(u_A) = f(v_A)$. Since f was arbitrary, $u_A \sim_{\mathcal{F}_A} v_A$. Conversely, $u_A \sim_{\mathcal{F}_A} v_A$ gives $c_{f,g}(u, w) = c_{f,g}(v, w)$ for all g, w , so $L(u) = L(v)$.

(iv). Fix $g \in \mathcal{F}_B$; use constant functions $f_t \in \mathcal{F}_A$ and the identity $c_{f_t,g}(w, u) = \mathbf{1}[t \geq g(u_B)]$. The same threshold argument as in (iii) shows that $u \beta v$ implies $u_B \sim_{\mathcal{F}_B} v_B$. Conversely, if $u_B \sim_{\mathcal{F}_B} v_B$, then $c_{f,g}(w, u) = c_{f,g}(w, v)$ for all f, w , so $R(u) = R(v)$.

(v). Immediate from (iii)–(iv). When $\mathcal{F}_A$ separates Y_A , $\sim_{\mathcal{F}_A}$ is equality, so $U/\alpha \cong Y_A$; similarly for U/β . $\square$

5 Main Results

5.1 Classical Hamiltonian systems

Theorem 5.1 (Classical equivalence). *Let X_A and X_B be the phase spaces of subsystems A and B . Let $\mathcal{F}_A$ and $\mathcal{F}_B$ be separating families of observables on X_A and X_B respectively, each containing all constant observables, and let $\mathcal{C} = \mathcal{C}_{cs}$. Assume every subsystem state is realized by some joint state: $\text{pr}_A(U) = X_A$ and $\text{pr}_B(U) = X_B$.*

A two-subsystem classical system with joint state space $U \subseteq X_A \times X_B$ and comparison predicates $\mathcal{C} \upharpoonright U$ is rectangularly complete if and only if $U = X_A \times X_B$, i.e., the system satisfies the full-product condition in Definition 2.1(i).

Proof. ($\Leftarrow$) *Full product implies rectangularly complete.* Suppose $U = X_A \times X_B$. By Lemma 4.6(v), $U/\alpha \cong X_A$ and $U/\beta \cong X_B$. Under these identifications $\Theta: U \rightarrow X_A \times X_B$ sends $(u_A, u_B) \mapsto (u_A, u_B)$, which is the identity map and hence a bijection. Rectangular completeness holds.

($\Rightarrow$) *Rectangularly complete implies full product.* Suppose $(U, \mathcal{C} \upharpoonright U)$ is rectangularly complete; suppose for contradiction that $U \subsetneq X_A \times X_B$. Pick $(a, b) \in (X_A \times X_B) \setminus U$. By the surjectivity assumptions choose $(a, b') \in U$ and $(a', b) \in U$. Set

$$A_a := [(a, b')]_\alpha \in U/\alpha, \quad B_b := [(a', b)]_\beta \in U/\beta.$$

By Lemma 4.6(iii)–(iv) these classes depend only on a and b , not on a' or b' . If some $u = (u_A, u_B) \in U$ realized the pair (A_a, B_b) , then the same lemma would give $u_A = a$

and $u_B = b$, hence $u = (a, b) \notin U$, a contradiction. Thus (A_a, B_b) is an unrealized pair in $U/\alpha \times U/\beta$, contradicting rectangular completeness. Hence $U = X_A \times X_B$. $\square$

Remark 5.2 (Exact equivalence, not mere implication). The theorem establishes a biconditional between rectangular completeness and the full-product condition in the standard classical definition. The surjectivity assumptions are normalization conditions, not extra closure hypotheses: they identify X_A and X_B as the subsystem state spaces actually present in the system under study. With that understood, the two definitions express the same structural content in different languages.

Example 5.3 (Constrained system: the biconditional is non-trivial). Consider two classical particles A and B moving on the real line and joined by a rigid rod of fixed length $L > 0$. The rod may be treated as part of a closed composite mechanical system. Relative to the two distinguished profiles A and B , however, it imposes a relation between the allowed subsystem coordinates. The rod forces $q_B - q_A = L$, so the effective joint state space is

$$U_{\text{rod}} = \{(q, q + L) : q \in \mathbb{R}\} \subsetneq \mathbb{R} \times \mathbb{R} = X_A \times X_B.$$

With cross-subsystem observables $f = g = \text{id}$, the α -congruence on U_{rod} identifies states by their A -position and β by their B -position, giving $U_{\text{rod}}/\alpha \cong \mathbb{R}$ and $U_{\text{rod}}/\beta \cong \mathbb{R}$. The canonical factor map is $\Theta(q, q + L) = (q, q + L) \in \mathbb{R} \times \mathbb{R}$, whose image is the affine diagonal $\{(q, q + L) : q \in \mathbb{R}\}$. Any pair (q_A, q_B) with $q_B - q_A \neq L$ is unrealized; for instance, the pair $(0, 0)$ has no representative in U_{rod} . Rectangular completeness fails, and Theorem 5.1 establishes $U_{\text{rod}} \subsetneq X_A \times X_B$. The conclusion is not that the composite rod system is open in the ordinary mechanical sense. It is that, after A and B have been selected as independently meaningful subsystem profiles, the rod supplies additional relational data that restricts which A -profiles may be paired with which B -profiles. In the language of [1, Def. 10.1], the unrealized (α, β) pairs are profile-level witnesses that the chosen two-profile description is not rectangularly complete.

5.2 Thermodynamic systems

Corollary 5.4 (Thermodynamic equivalence). *Let X_A^{th} and X_B^{th} be the thermodynamic state spaces of components A and B , let $\mathcal{F}_A^{\text{th}}$ and $\mathcal{F}_B^{\text{th}}$ be separating families of thermodynamic observables (internal energy, entropy, volume, particle numbers, etc.) each containing all constant observables, and let $\mathcal{C}_{\text{cs}}^{\text{th}}$ be the induced cross-subsystem family.*

A two-component thermodynamic system with preparation state space $U^{\text{th}} \subseteq X_A^{\text{th}} \times X_B^{\text{th}}$, satisfying $\text{pr}_A(U^{\text{th}}) = X_A^{\text{th}}$ and $\text{pr}_B(U^{\text{th}}) = X_B^{\text{th}}$, is rectangularly complete (with respect to $\mathcal{C}_{\text{cs}}^{\text{th}}$) if and only if $U^{\text{th}} = X_A^{\text{th}} \times X_B^{\text{th}}$, i.e., its preparation state space has the full-product form associated here with a closed or isolated two-component system. In

particular, the equilibrium manifold of any two-component system placed in thermal or mechanical contact is never rectangularly complete.

Proof. Theorem 5.1 applies with $X_A, X_B, \mathcal{F}_A, \mathcal{F}_B$ replaced by $X_A^{\text{th}}, X_B^{\text{th}}, \mathcal{F}_A^{\text{th}}, \mathcal{F}_B^{\text{th}}$. The surjectivity hypotheses play the same normalizing role as in the classical theorem. The “in particular” clause holds because any equilibrium constraint (equal temperatures, pressures, or chemical potentials) restricts the preparation state space to a proper subset of the product, which by the necessity direction leaves profile-class pairs unrealized. $\square$

Remark 5.5 (Preparation versus equilibrium). The corollary applies to the preparation state space, not the equilibrium manifold. The preparation state space of a closed or isolated system is the full product: any macrostate of A and any macrostate of B can be independently prepared before being brought together. Once a contact condition—for example a diathermal wall, a movable partition, or a semipermeable membrane—is imposed, it restricts which macrostate pairings are represented on the equilibrium manifold. The resulting manifold need not be rectangularly complete, as the following example makes precise.

Example 5.6 (Oxygen and nitrogen: preparation versus contact). Let A be oxygen and B nitrogen, each in its own insulated container with macrostate (T, V, N) . Before any contact, any combination of macrostates is independently preparable: $U^{\text{th}} = X_A^{\text{th}} \times X_B^{\text{th}}$ is the full product, and Corollary 5.4 confirms rectangular completeness.

Insert a rigid diathermal wall (thermal contact, no matter or volume exchange). The equilibrium manifold is

$$U_{\text{eq}} = \{(T, V_A, N_A, T, V_B, N_B) : T, V_i > 0, N_i \in \mathbb{N}\} \subsetneq X_A^{\text{th}} \times X_B^{\text{th}}.$$

With the cross-subsystem predicate comparing T_A to T_B , the α -classes are indexed by temperature of A and the β -classes by temperature of B . Any profile-class pair (T_A, T_B) with $T_A \neq T_B$ has no representative in U_{eq} ; rectangular completeness fails. The wall is not a problem for ordinary thermodynamic terminology; it is the physical condition selecting a contact equilibrium. Rectangular completeness records the narrower profile fact: after the equilibrium condition is imposed, the manifold no longer contains every independently preparable pair of component macrostates.

5.3 Quantum systems

The Hilbert-space tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ contains entangled states not of the product form $\psi_A \otimes \psi_B$. We treat the product-state sector and the full state space separately.

5.3.1 Product-state sector

Let $S_A := \{\psi_A : \|\psi_A\| = 1\}/U(1)$ and $S_B := \{\psi_B : \|\psi_B\| = 1\}/U(1)$ be the projective state spaces of $\mathcal{H}_A$ and $\mathcal{H}_B$, and let

$$U_{\text{prod}} := \{[\psi_A \otimes \psi_B] : [\psi_A] \in S_A, [\psi_B] \in S_B\} \cong S_A \times S_B.$$

Let $\mathcal{F}_A^Q$ be the family of pure-state expectation-value functionals $[\psi_A] \mapsto \langle \psi_A, O_A \psi_A \rangle$ as O_A ranges over bounded self-adjoint operators on $\mathcal{H}_A$, and $\mathcal{F}_B^Q$ the analogous family for $\mathcal{H}_B$. These families separate their respective projective state spaces by the completeness of the observable algebra, and contain constant functionals via scalar operators λI .

Theorem 5.7 (Quantum product-sector equivalence). *Let $\mathcal{C}_{\text{cs}}^Q$ be the cross-subsystem family on $U_{\text{prod}} \cong S_A \times S_B$ generated by $(\mathcal{F}_A^Q, \mathcal{F}_B^Q)$. Then $(U_{\text{prod}}, \mathcal{C}_{\text{cs}}^Q)$ is rectangularly complete, and the canonical decomposition recovers $X_A \cong S_A$, $X_B \cong S_B$.*

Proof. $U_{\text{prod}} = S_A \times S_B$ is a full product with separating observable families, so Theorem 5.1 applies directly with S_A and S_B in place of X_A and X_B . $\square$

5.3.2 Entanglement and the boundary of product-profile data

Proposition 5.8 (Marginal-profile data do not give rectangular completeness on the full quantum state space). *Let $U_Q = \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$. For self-adjoint O_A on $\mathcal{H}_A$ and O_B on $\mathcal{H}_B$, define*

$$c_{O_A, O_B}(\rho, \sigma) := \mathbf{1}[\text{Tr}(O_A \rho_A) \geq \text{Tr}(O_B \sigma_B)],$$

where $\rho_A = \text{Tr}_B \rho$ and $\sigma_B = \text{Tr}_A \sigma$. Let $\mathcal{C}$ be the family of all such predicates. Then $(U_Q, \mathcal{C})$ is not rectangularly complete whenever $\dim \mathcal{H}_A \geq 2$ and $\dim \mathcal{H}_B \geq 2$.

Proof. If $\rho_A = \sigma_A$ then $c_{O_A, O_B}(\rho, \tau) = c_{O_A, O_B}(\sigma, \tau)$ for all O_A, O_B, τ , so $\rho \alpha \sigma$. Conversely, if $\rho \alpha \sigma$, choose $O_B = tI$ to obtain $\mathbf{1}[\text{Tr}(O_A \rho_A) \geq t] = \mathbf{1}[\text{Tr}(O_A \sigma_A) \geq t]$ for all t . If $\text{Tr}(O_A \rho_A) \neq \text{Tr}(O_A \sigma_A)$, choosing t strictly between the two values gives different indicators, a contradiction. Hence $\text{Tr}(O_A \rho_A) = \text{Tr}(O_A \sigma_A)$ for every self-adjoint O_A , and therefore $\rho_A = \sigma_A$. So $\rho \alpha \sigma \iff \rho_A = \sigma_A$; by symmetry, $\rho \beta \sigma \iff \rho_B = \sigma_B$. The canonical factor map $\Theta: \rho \mapsto (\rho_A, \rho_B)$ is not injective: the Bell state

$$\rho := |\Phi^+\rangle\langle\Phi^+|, \quad |\Phi^+\rangle := \frac{|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B}{\sqrt{2}},$$

and the product state $\sigma := \frac{1}{2}(|0\rangle_A\langle 0| + |1\rangle_A\langle 1|) \otimes \frac{1}{2}(|0\rangle_B\langle 0| + |1\rangle_B\langle 1|)$ satisfy $\rho_A = \sigma_A = \frac{1}{2}I_A$ and $\rho_B = \sigma_B = \frac{1}{2}I_B$, yet $\rho \neq \sigma$. Hence Θ is not injective and rectangular completeness fails. $\square$

Remark 5.9 (Entanglement as correlation structure). Proposition 5.8 delineates the boundary of what the present product-profile comparison captures. The product-state sector is rectangularly complete because each state is determined by an A profile together with a B profile. The full quantum state space contains additional global correlation structure, including entanglement, not fixed by the pair of marginals. The broader theoretical development in [2] treats entanglement-type enrichment as part of the morphism-level transport structure identified by the two-locus exhaustion theorem (Theorem 9.10 of [1]).

6 Strict Generality of Rectangular Completeness

The results of Section 5 show that rectangular completeness contains the standard classical and thermodynamic full-product conditions considered here, together with the quantum product-sector case. We now show that it is *strictly* more general.

Example 6.1 (Finite discrete closed world). Let $X_A = \{a_1, a_2\}$, $X_B = \{b_1, b_2\}$, $U = X_A \times X_B$. Define

$$\begin{aligned} c_1((u_A, u_B), (v_A, v_B)) &= \mathbf{1}[u_A = a_1] \cdot \mathbf{1}[v_B = b_2], \\ c_2((u_A, u_B), (v_A, v_B)) &= \mathbf{1}[u_A = a_2] \cdot \mathbf{1}[v_B = b_1]. \end{aligned}$$

These are cross-subsystem predicates with $f_1 = \mathbf{1}_{a_1}$, $g_1 = \mathbf{1}_{b_2}$, $f_2 = \mathbf{1}_{a_2}$, $g_2 = \mathbf{1}_{b_1}$. The left profiles distinguish a_1 from a_2 and the right profiles distinguish b_1 from b_2 ; hence α identifies states by their A -component and β by their B -component. All four (α, β) pairs are realized exactly once. The comparison world $(U, \{c_1, c_2\})$ is rectangularly complete; it has no topology, no metric, no Hamiltonian, and no dynamics, and corresponds to no classical, thermodynamic, or quantum system.

Proposition 6.2 (Strict inclusion). *Let $\mathcal{RC}$ denote the class of all rectangularly complete comparison worlds, and let $\mathcal{PC}$ denote the class of two-subsystem systems satisfying the classical or thermodynamic full-product conditions isolated above, together with the product-state sectors of standardly closed bipartite quantum systems. When $\mathcal{PC}$ is embedded into $\mathcal{RC}$ via its natural comparison world (Section 4):*

- (i) $\mathcal{PC} \subseteq \mathcal{RC}$;
- (ii) $\mathcal{PC} \subsetneq \mathcal{RC}$.

Proof. (i) is the content of Theorem 5.1, Corollary 5.4, and Theorem 5.7. (ii) follows from Example 6.1, which is in $\mathcal{RC}$ but has no physical dynamics and therefore is not in $\mathcal{PC}$. $\square$

Remark 6.3 (What strict generality means). From the perspective of physics, rectangular completeness applies where standard physical notions are not defined: finite

relational systems, combinatorial state spaces, logical models. It isolates the structure of closure from the physical machinery surrounding it. From the perspective of mathematics, rectangular completeness is the unique minimal condition (Theorem 3.5) forcing the canonical product decomposition $U \cong X_A \times X_B$ from pure comparison data. The physical definitions of closure are *realizations* of that structural condition in specific contexts.

7 Conclusion

We have proved that the comparison-world condition of rectangular completeness stands in the following relation to the product-structure component of standard two-subsystem formalisms:

- (a) *Classical Hamiltonian systems.* Rectangular completeness is exactly equivalent to the full-product component of standard closure. The theorem is a biconditional after the separating-observable and realization normalizations have been fixed. (Theorem 5.1, Example 5.3.)
- (b) *Thermodynamic systems.* The same equivalence holds on preparation state spaces. The equilibrium manifold of a system in thermal or mechanical contact is never rectangularly complete. (Corollary 5.4, Example 5.6.)
- (c) *Quantum product sector.* Rectangular completeness captures the product-state sector of a closed bipartite quantum system. Marginal-based comparison data do not separate the full quantum state space; entanglement supplies correlation data not determined by subsystem marginal profiles. (Theorem 5.7, Proposition 5.8.)
- (d) *Strict generality.* Rectangular completeness applies to comparison worlds with no dynamics, metric, or topology, forming a class strictly larger than the physical sectors considered here. (Proposition 6.2.)

The composite result is therefore precise: *rectangular completeness is exactly equivalent to the structural full-product condition in the classical and thermodynamic cases, exactly equivalent on the quantum product-state sector, and the minimal structural extension of that condition across the full class of comparison worlds.*

These results justify the use of rectangular completeness as the comparison-world closure criterion in [1]: every two-subsystem classical or thermodynamic system satisfying the full-product condition considered here, and every quantum product sector of a standardly closed bipartite system, induces a rectangularly complete comparison world. Rectangular completeness is the minimal relational condition with that property.

Broader implications. The exact structural equivalence in the classical case means that the results of [1]—quotient semantics (Thm. 3.2), the failure of subsystem attribution (Thm. 7.1), and the two-locus exhaustion theorem (Thm. 9.10)—apply to

every classical two-subsystem system whose joint state space is the full product, without additional product-availability assumptions. The thermodynamic case shows the same structural picture governs preparation spaces before equilibrium constraints are imposed. The quantum case shows these results apply on the product-state sector of closed bipartite quantum systems, with global correlation structure, including entanglement, representing the residual enrichment identified by the two-locus theorem.

Statements and Declarations

Competing Interests The author declares that there are no competing interests.

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